

GoodJob: Mining Job Market Intelligence from Web Job Listings

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1. Problem Statement

The technology job market evolves at an accelerating pace, with rapidly shifting skill requirements, emerging roles, and complex interdependencies between competencies. Job seekers and career switchers face two fundamental challenges:

- Information overload. Thousands of job postings across multiple platforms make it difficult to synthesize a coherent picture of which skills are genuinely in demand, how they relate to one another, and which career directions offer the best growth trajectory.
- Lack of structured career intelligence. Existing job platforms present listings as isolated documents rather than connected data points. There is no systematic way to explore skill clusters, identify prerequisite relationships, visualize career progression paths, or receive personalized learning recommendations grounded in real-time market data.

Our goal is to design and build a web mining pipeline that transforms raw job listings into an interactive, queryable knowledge graph of skills, roles, and career pathways.

2. Approach

Stage 1: Web Scraping & Ingestion

We will scrape job listings from major tech company career portals using Playwright-based headless browser automation. The scraper will iterate over paginated search results and extract job title, department, full description, and requirements text for each listing. Collected data will be stored in structured JSON format for downstream processing.

Stage 2: Text Preprocessing & Entity Extraction

Raw text fields will be cleaned through Unicode normalization and stopword removal. A curated domain dictionary covering approximately 80 tech skills (programming languages, frameworks, tools, and core concepts) will be used for rule-based named entity recognition. Each job listing will be tagged with the set of skills mentioned in its title, description, or requirements section.

Stage 3: Job Classification

A rule-based classifier will assign each listing to one of several functional categories. Classification will use weighted keyword matching on both title and body text, supplemented by manual overrides for ambiguous edge cases.

Stage 4: Graph Construction & Analytics

We will build a heterogeneous knowledge graph with three node types - category, job, and skill - connected by category-to-job and job-to-skill edges. Mining operations will include: (a) skill frequency ranking per category, (b) skill triple co-occurrence analysis to identify high-demand skill combinations, and (c) career route evaluation via Directed Acyclic Graph (DAG) traversal.

Stage 5: Visualization & Interaction

The knowledge graph will be rendered as an interactive visual scene, enabling zoom and click-to-explore behavior on any node, along with job search functionality. A side panel will display detailed information including skill frequency distributions (pie charts), ranked skill lists, skill combination tables, and related job listings. A DAG-based skill planner will allow users to mark mastered skills and compute readiness scores for predefined career paths.

Stage 6: Personalized Page

We will implement resume-job matching via skill vector similarity, generate personalized learning path recommendations based on identified skill gaps, and build user profile pages with goal tracking and progress visualization to support long-term career development.

3. Datasets

We plan to use publicly available job postings from major internet companies such as ByteDance, Meituan, and JD.com, expecting to collect several thousand job listings in total. Each record will include: job title (raw and cleaned), department/section, full HTML description text, detailed requirements section, and posting URL. The data will be collected in July 2026 and will represent a snapshot of Chinese technology companies' hiring demand at that time.

4. Expected Outcomes

- A reusable web scraping and text mining pipeline for job market intelligence
- An interactive knowledge graph visualizing relationships between job roles and required skills
- A skill combination mining module that identifies high-demand skill triples across career domains
- A career route evaluator that computes skill readiness against predefined career paths
- A live web application demonstrating all components